

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-4/T-2 CSE 471: Machine Learning

Time: 25 minutes

Marks: 20 (+ 2 bonus)

Student Name: _____

Student No: _____

1. For each of the following propositions, identify each one as TRUE or FALSE. Write your answer in the leftmost column. ($1.5 \times 4 = 6$)

	The number of parameters in a convolutional layer is independent of the input image height and width.
	If we double the number of channels in each layer of a CNN, we always double the number of parameters in the network.
	ResNet’s skip connections allow gradients to flow directly to earlier layers without passing through the weight layers.
	Summing up cross entropy for each pixel is a valid loss function for semantic segmentation.

2. Consider the following input minibatch to a fully connected layer. Please note that the minibatch has four data points and each data point consists of three features.

$x_0 = [+7 \quad +2 \quad +3]$ $x_1 = [+2 \quad -4 \quad -1]$

$x_2 = [-5 \quad +2 \quad +2]$ $x_3 = [+3 \quad +3 \quad -3]$

- (a) Apply batch normalization. (3)
- (b) Apply layer normalization. (3)

[N.B. Just detail down the processes. You don’t need to do detailed floating point calculations.]

3. Assume you have an image of resolution 512×512 with three channels. You want to convert it into an image of the same resolution with thirty channels. How can you apply convolution to do the job? (2)
4. Illustrate the usage of stride in up convolutions with an appropriate example. (3)
5. Explain the concept of fine-tuning in the context of computer vision with an appropriate example. (3)
6. Mention two use cases of `paperswithcode.com` (preferably those mentioned in the class). (2)